Evaluating Multilingual Reasoning through a Chain-of-Thought Lens

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Abstract

Does asking a language model to think step by step help it reason about dates and times, and does the answer change with language? We test Sarvam-105B on 6,750 questions from the Temporal Reasoning Dataset in English and Hindi, with and without distractor sentences. Chain-of-thought (CoT) raises English accuracy by 29.5 points and Hindi accuracy by 4.7, so the gap between the languages grows from about 4 points to about 29. Asking for English reasoning on the same Hindi questions closes about four-fifths of this gap. A two-turn cross-lingual prompt closes about a third, and Hinglish reasoning closes none of it on Hindi questions. Hidden thinking mode almost closes the gap, although a third of its steps on Hindi direct prompts argue about the output format. The Hindi losses appear to come from methods the model uses mainly when it writes Hindi, such as subtracting dates by counting forward. Traces that differ more go with answers that disagree, but much of this link seems to come from the answers themselves. Our trace experiments could not show that a distractor caused an error. All results come from one model and one prompt per condition.

1 Introduction

Chain-of-thought (CoT) prompting asks a language model to write intermediate steps before it answers, which can improve multi-step reasoning (Wei et al., 2022; Kojima et al., 2022). We call this text a "reasoning trace", and we do not assume that it records the model's real computation (Turpin et al., 2023). Language may matter too: the MGSM study found English steps as good as or better than steps in the question's language (Shi et al., 2023b). So if we ask a model to reason in Hindi, does it reason as well as it does in English, and does it get there the same way?

We study this with the Temporal Reasoning Dataset (TRD) of Mazzia et al. (2026), which turns CoT off and leaves its effect to future work. TRD can add a distractor sentence to any question. Some distractors concern time ("I always get confused with months that have 31 days"), and others do not ("My brother bought a motorcycle"). This gives a direct test:

> If a model can ignore the motorcycle when it answers directly, does thinking step by step give the motorcycle more chances to enter its reasoning and change the answer?

We use Sarvam-105B, whose developer reports a focus on the ten most-spoken Indian languages (Sarvam AI, 2026).

Our contributions are: (1) to our knowledge, the first CoT evaluation on TRD, split by distractor condition; (2) a split of the English-Hindi gap into reasoning and question language; (3) an error analysis of specific methods; (4) a regression of disagreement on trace divergence, with a check of how much comes from the answers; and (5) deletion and resampling experiments on distractor mentions. We recomputed the reported numbers from raw responses and corrected our mid report where it overclaimed.

2 Problem Statement

We work with one model and with TRD problems whose answers have a fixed form, such as a date or a weekday, so responses can be scored automatically. We vary the question language, the distractor (none, time-related or unrelated), the prompt, and the language the model is asked to reason in. A *direct* prompt asks only for the answer. A *CoT* prompt asks for step-by-step reasoning.

All differences are in percentage points. The CoT *gain* is CoT accuracy minus direct accuracy. The *gap* is English accuracy minus Hindi accuracy. A distractor's *drop* is accuracy without it minus accuracy with it. The CoT *penalty* is the drop under

Work	Mul.	Tem.	CoT	Dis.	Tra.
Wei et al. (2022)	–	○	✓	–	–
Shi et al. (2023a)	–	–	✓	✓	–
Shi et al. (2023b)	✓	–	✓	–	–
Qin et al. (2023)	✓	–	✓	–	–
Qi et al. (2025)	✓	–	✓	–	○
Golovneva et al. (2023)	–	–	✓	–	✓
Chu et al. (2024)	–	✓	✓	–	–
Mazzia et al. (2026)	✓	✓	–	✓	–
Ours	✓	✓	✓	✓	✓

Table 1: Prior work by multilinguality (Mul.), temporal focus (Tem.), CoT, distractors (Dis.) and trace analysis (Tra.); ○ means partly covered.

CoT minus the drop under direct answering; a positive value suggests that thinking out loud gives the distractor room to do harm. Square brackets give 95% intervals. We ask six questions:

RQ1 Does CoT help, and does that depend on language and difficulty?

RQ2 Does CoT make distractors more harmful?

RQ3 How much of the English-Hindi gap follows the language of reasoning rather than the language of the question?

RQ4 Does asking for Hinglish reasoning help?

RQ5 Does divergence between English and Hindi traces track whether the two languages disagree?

RQ6 Can we say that distractors in CoT traces cause errors?

3 Related Work

CoT and when it stops helping. Wei et al. (2022) introduced CoT with worked examples, and Kojima et al. (2022) showed that a trigger phrase alone can work, which suits TRD because it has no worked solutions. Sprague et al. (2025) find that CoT helps mainly on math and symbolic problems, so date arithmetic might be expected to benefit. It can also lower performance on some tasks (Liu et al., 2025).

Distractors. GSM-IC adds one irrelevant sentence to GSM8K problems, and accuracy falls under several prompting methods (Shi et al., 2023a). GSM-Symbolic shows sensitivity to clauses that look relevant but are not (Mirzadeh et al., 2025). Neither compares CoT on and off under the same distractor, and both work in English.

Language of reasoning. MGSM compares native and English CoT on ten languages that do not include Hindi (Shi et al., 2023b). Cross-lingual prompting (CLP) adds an understanding turn before the solving turn (Qin et al., 2023). Thinking models often revert to English, and forcing the user's language can cost accuracy (Qi et al., 2025; Yong et al., 2025). Some models may process input in an English-leaning way (Wendler et al., 2024; Zhao et al., 2024). Hinglish is an established code-switching pair (Khanuja et al., 2020; Aguilar et al., 2020), and code-mixed examples can help on some tasks (Shankar et al., 2024), though these studies use code-mixed input and do not test generated reasoning.

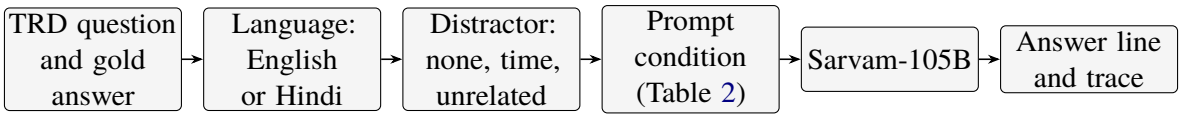
Traces and faithfulness. Faithfulness can be tested by intervening on a trace (Lanham et al., 2023). Thought Anchors resamples continuations to estimate the effect of each sentence (Bogdan et al., 2025), and Thought Branches argues that hand edits are off-policy and may understate effects (Macar et al., 2026). Step quality has been scored directly (Golovneva et al., 2023; Prasad et al., 2023); our divergence score is structural and does not assess validity.

The gap. TimeBench and TRAM evaluate temporal reasoning mostly in English (Chu et al., 2024; Wang and Zhao, 2024). As Table 1 shows, we did not find a study that crosses CoT, distractors and reasoning language on the same problems, though our search was not systematic.

4 Experimental Setup

Data. Figure 1 shows the pipeline. TRD has nine tasks: date addition, subtraction and duration, time addition, subtraction and duration, date recurrence, interval date (the start and end of next week) and day of week. We generated 150 questions per task at each of five difficulty levels, 6,750 in all, in English and Hindi with matching IDs. Each question appears three times: with no distractor, with a time-related one and with an unrelated one, drawn from pools of 35 and 43 sentence pairs. We recomputed all 13,500 gold answers with Python's `datetime`, and all match.

Response sets. Table 2 lists the sets. Unless stated, runs use temperature 0, thinking off, a 4,096-token limit and a fixed answer line. Thinking-on uses `reasoning_effort` low and 16,000 tokens, on 10 questions per cell. en-CoT keeps the Hindi



English: Today is 2027-05-16, and I have a recurrence every 13 days. Without counting today, what is the date of the 7 occurrence? **Gold:** 2027-08-15.
Hindi: आज 2027-05-16 है, और मेरे पास हर 13 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 7वीं आवृत्ति की तिथि क्या है? (*āj 2027-05-16 hai, aur mere pās har 13 dinoṃ par ek punarāvṛtti hai. āj ko choṛkar, 7vīṃ āvṛtti kī tithi kyā hai?*, the same question).
Time-related distractor: The last time I had a weekly commitment was in January. / आखिरी साप्ताहिक प्रतिबद्धता जनवरी में थी। (*ākhirī sāptāhik pratibaddhatā janvarī meṃ thī*, 'The last weekly commitment was in January.')

Condition	No distractor	Time-related	Condition	No distractor	Time-related
English direct	2027-06-05 ×	2027-04-19 ×	en-CoT (Hindi Q.)	2027-08-15 ✓	cut off
English CoT	2027-08-15 ✓	cut off	CLP (Hindi Q.)	2027-08-02 ×	2027-08-02 ×
Hindi direct	2027-07-27 ×	2027-03-06 ×	Hinglish (English Q.)	2027-08-02 ×	2027-08-01 ×
Hindi CoT	2027-08-02 ×	2027-04-20 ×	Hinglish (Hindi Q.)	2027-08-02 ×	2027-07-06 ×

Figure 1: The pipeline, with real responses to one question (long, date recurrence, ID 12), picked to illustrate the design. The answer 2027-08-02 adds six intervals where seven are needed. With the January sentence, English CoT and en-CoT keep trying to fit January into the question until the token limit.

Set	Prompts	Q.	Resp.
Standard	Direct, CoT	6,750	81,000
Thinking-on	Direct, CoT	450	5,400
en-CoT	Hindi Q., English CoT	2,250	6,750
CLP	Hindi Q., two turns	2,250	6,750
Hinglish	Both Q., mixed CoT	2,250	13,500

Table 2: Response sets, in the order of the results (Q. is questions, Resp. is responses). Trace interventions add 13,664 resampled continuations.

question and asks for English reasoning, a zero-shot version of Shi et al. (2023b). CLP first asks the model to understand the Hindi problem in English and then to solve it (Qin et al., 2023). We fix temperature at 0, add a rule to write in English, and skip their voting variant. Hinglish asks for reasoning that mixes Hindi and English in Roman script.

Scoring. A response with no answer line usually hit the token limit. *Answered* accuracy leaves such responses out, and our figures use it. *Strict* accuracy counts them as wrong, which is closer to TRD's own scoring. English CoT has the most cut-offs among thinking-off conditions, so answered accuracy flatters it, and we give both for headline results.

Trace measures. We record each trace's length, whether it has a verification phrase (regex), and whether it mentions the distractor, compared with a placebo rate on the same question without the distractor. The target-language ratio (TLR) is the share of Devanagari words among Devanagari and Latin-script words. A script labels every wrong response by how far off it is and the kind of mistake. This covers 30,421 of the 93,150 responses in the first three sets, with cut-offs counted as wrong.

Comparing traces. A regex parser turns each trace into a sequence of steps with operands, such as `SUB 2025-05-26 4` for "4 days before 26 May 2025". Labels include `CHECK`, `REVISE` and `FORMAT` (arguing about the output format or language). The reasoning divergence score (RDS) between two traces is the Levenshtein distance (Levenshtein, 1966) between their sequences divided by the longer length. A score of 0 means the same steps, and 1 means no step in common. The parser covers 95 to 96% of CoT segments on standard replies and 86 to 88% on thinking-mode reasoning. On 240 tokens read by one LLM annotator, it gets the operation right 85% of the time and the operation with its arguments 77%.

Statistics. Accuracy intervals are bootstrap intervals (Berg-Kirkpatrick et al., 2012) that resample questions, since a question appears in three distrac-

	Short answer	Section
RQ1	Yes for English (+29.5 points); little for Hindi (+4.7).	5.1.2
RQ2	Slightly in Hindi, largely through one sentence (penalty 2.2).	5.1.2
RQ3	Most of it, descriptively: 23.5 of the 29.5-point gap.	5.3
RQ4	No: −0.0 on Hindi questions, −18.0 on English ones.	5.5
RQ5	Yes as an association (OR 4.93), though much of it seems to come from the answers.	5.1.3
RQ6	No. Removing the distractor helps a little (+5.3).	5.6

Table 3: Short answers to the research questions.

tor conditions. Task-level tests use exact McNemar with Holm correction over nine tasks (McNemar, 1947; Holm, 1979; Dror et al., 2018), and mention odds ratios (ORs) are Mantel-Haenszel (Mantel and Haenszel, 1959).

5 Results

We report each response set in turn, with the main finding in bold. Table 3 answers each research question, and Table 4 compares the sets on the same 2,250 questions.

5.1 Standard Set

5.1.1 Direct Answers

Without reasoning, both languages score low and close together: 59.0% in English and 54.7% in Hindi, a gap of 4.3 points.

Accuracy. Day-of-week answers look like guesses. They are wrong 85% of the time (3,839 of 4,500), close to the rate expected by chance (85.7%). They also favour one weekday whatever the date: Friday for 34.4% of English answers and Sunday for 58.7% of Hindi answers. Hindi direct is also wrong on 97% of recurrences (2,180 of 2,250), against 80% in English.

Errors. Of wrong date answers, 35% in English and 30% in Hindi are off by exactly 10, 20 or 30 days. Asked for 36 days after 2029-03-06, the model says 2029-04-01 (gold 2029-04-11). This falls to 3% under English CoT. In time tasks the minutes are usually right and the hours wrong: 65% of wrong English and 57% of wrong Hindi time answers are off by whole hours. On interval date, 22% of wrong English answers give this week instead of next.

Distractors. Distractors appear to change which wrong answer the model gives. Wrong Hindi answers reuse a number, month or weekday from a time-related distractor 11.3% of the time, against a placebo of 4.1% ($p < 0.001$). Wrong English answers do so 6.1% of the time, against 3.4% ($p = 0.004$). Three recurrence sentences naming February or January supply 78% of the Hindi reuse.

5.1.2 Chain-of-Thought

CoT adds 29.5 points in English and 4.7 in Hindi, so the gap grows from 4.3 points to 29.1 (strict 26.7). The gap is similar with and without distractors (Table 5), so pooling does not appear to create it.

Accuracy. English CoT is never significantly below direct answering on any task at any level. Hindi CoT can be: even with no distractor, it lowers date subtraction by 19.9 points [16.4, 23.4] and interval date by 23.3 [19.0, 27.7]. The gap widens with difficulty, from 24.0 points on the shortest questions to 34.9 on the longest, but a logistic trend test is not significant ($p = 0.21$).

Distractors. On average the CoT penalty is small. A time-related distractor lowers Hindi CoT accuracy by 2.6 points [1.3, 3.8], which is 2.2 more than it lowers Hindi direct answers. With the 45 task-sentence units as the sample, the interval reaches zero (+2.3 [−0.1, 4.6]), so the penalty may not extend to new sentences. Whether Hindi is penalised more than English depends on how unanswered responses are scored (Appendix B), so we withdraw our mid-report claim of fragility only in Hindi.

The largest effect among 90 sentences, found post hoc, comes from "The last time I had a weekly commitment was in January" (Figure 1). On date recurrence, English CoT runs to the token limit on 92 of 134 questions (4 of 134 without it). Strict English accuracy falls from 88.8% to 19.4%. Hindi has no cut-offs but falls from 26.9% to 2.2%. This sentence competes with the question's start date, so it may not count as irrelevant in the GSM-IC sense.

Hindi traces mention distractors more: 25.0% against 8.5% for time-related sentences, with a near-zero placebo on a post hoc word list. Traces that mention a time-related distractor are less often cor-

	English Q.		Hindi Q.				Hinglish prompt	
	Direct	CoT	Direct	CoT	CLP	en-CoT	English Q.	Hindi Q.
Date addition	85.6	98.0	82.8	85.1	92.7	95.7	87.3	85.9
Date subtraction	86.0	95.2	84.8	66.0	86.9	94.0	70.7	71.0
Date duration	84.0	99.3	78.9	88.2	92.7	97.7	92.9	82.8
Date recurrence	22.1	90.3	2.8	20.1	26.9	70.3	53.8	18.9
Day of week	14.9	69.5	15.2	21.0	31.8	52.5	34.7	23.5
Interval date	43.2	50.7	46.8	23.9	35.3	44.1	38.5	23.6
Time addition	70.5	95.9	75.6	73.5	88.2	92.6	82.9	72.3
Time subtraction	64.7	97.7	64.9	75.2	90.1	95.7	85.2	74.2
Time duration	65.7	98.9	44.7	80.1	86.7	98.1	90.0	80.9
Short	77.8	93.2	73.9	69.5	77.2	83.0	82.9	66.5
Medium	69.6	91.6	65.3	65.4	75.7	84.9	78.8	67.4
Long	61.2	89.3	57.9	60.4	71.3	84.4	70.4	60.9
Very long	54.3	87.2	48.4	53.7	65.8	83.5	64.0	53.7
Very very long	35.5	82.9	30.3	47.9	61.5	78.5	58.2	48.3
No distractor	60.6	89.3	56.0	60.2	69.3	84.1	70.6	61.0
Time-related	59.9	88.2	54.9	58.9	69.3	81.0	69.7	59.1
Unrelated	58.6	89.2	54.6	59.1	72.3	83.6	72.3	58.0
All, answered	59.7	88.9	55.2	59.4	70.3	82.9	70.9	59.4
All, strict	59.6	86.0	55.2	59.0	69.9	80.8	70.3	59.1

Table 4: Answered accuracy (%) on the same 2,250 questions, by task, level and distractor condition. The last row counts unanswered responses as wrong. Differences in the text are computed before rounding, so they can differ from these cells by 0.1.

	Direct	CoT	Gain	Strict
English	59.0	88.5	+29.5	+26.8
Hindi	54.7	59.4	+4.7	+4.2
Distractor	None		Time	Unrelated
English	+28.8		+29.1	+30.7
Hindi	+5.6		+3.4	+5.0
Gap, CoT	+28.0		+29.7	+29.7
Gap, direct	+4.8		+4.0	+3.9

Table 5: CoT gain on all 6,750 questions, in points. Values are answered except the Strict column. Intervals on the gains are English [28.5, 30.6] and Hindi [3.6, 5.7]. Gains and gaps are computed before rounding.

rect (Hindi OR 0.77 [0.67, 0.89]). Some traces take the distractor literally. After "My watch runs slow", one Hindi trace reaches the right time, 23:28. It then writes (translated) "we cannot get 23:28 because my watch is slow" and answers 00:00.

Traces. Hindi CoT stays in Hindi: its TLR is 0.988, and only 2.14% of traces fall below 0.9. Those traces are more often correct within task and level (OR 2.08 [1.66, 2.60]), an observational link. Hindi traces are shorter (median 13 steps against 25), and 3.7% contain a `CHECK` or `REVISE` step, against 88% of English traces.

Errors. Day of week, interval date and date recurrence hold 64% of all wrong answers (19,583 of 30,421). They hold 78 to 88% when the model reasons in English, since it rarely gets plain arithmetic wrong (Table 6).

Hindi CoT's losses appear to come from methods seen mainly in Hindi traces. On date subtraction it often counts forward to the month end and then steps into the next month, so 13% of its wrong answers land on or after today. On interval date, a fifth of its wrong answers are "weeks" that are not seven days long. On time tasks it can wrap around midnight wrongly. For 19 hours 41 minutes after 10:46, one trace writes:

Error pattern	Condition	Share
Answer on or after today (date subtraction)	Hindi CoT	13% of 738
	English CoT	0% of 129
	en-CoT	0% of 45
Span not seven days (interval date)	Hindi CoT	21% of 1,720
	English CoT	6% of 1,046
	en-CoT	6% of 410
Counts backwards (date recurrence)	en-CoT	28% of 207
	Hindi CoT	9% of 1,758
	English CoT	3% of 234
One interval short (date recurrence)	Hindi CoT	35% of 1,758
	English CoT	21% of 234
One day late (day of week)	English CoT	58% of 584
	en-CoT	50% of 313

Table 6: Main CoT error patterns, as shares of wrong answers of the expected form on that task, over all levels and distractor conditions. Labels come from a script. Appendix A has examples.

> 24 - 19 = 5 घंटे। तो, 19 घंटे और 41 मिनट का मतलब है कि हमें 5 घंटे और 41 मिनट जोड़ने हैं।
> *24 - 19 = 5 ghaṇṭe. to, 19 ghaṇṭe aur 41 minaṭ kā matlab hai ki hameṃ 5 ghaṇṭe aur 41 minaṭ joṛne haiṃ.*
> '24 - 19 = 5 hours. So 19 hours and 41 minutes means we have to add 5 hours and 41 minutes.'

Its answer is 16:27 (gold 06:27). The error rate on time addition stays at 24 to 27% at every level, which suggests a fixed method more than a problem with larger numbers.

Other errors appear in both languages. On day of week, the most common error under English reasoning is exactly one day late, and the traces we read often count 1 January itself when adding up days. Recurrences often stop one interval short, apparently because the model counts today as the first occurrence. With 29 February in the span, English CoT on the longest questions is wrong 36% of the time against 9% without it. Yet 181 of its 186 wrong answers on such questions mention the leap year. Long re-checking loops go with failure: cut-off English CoT traces average 42 check or revise steps, against 8 for wrong traces that finish.

5.1.3 English and Hindi Traces Compared

More divergent traces go with disagreement, but much of this link seems to come from the final answers inside the score.

Regression. We call an English-Hindi pair of CoT traces *disagreeing* when exactly one language is right. A logistic regression predicts disagreement from RDS, controlling for task, difficulty, distractor and thinking mode, with standard errors clustered by question (20,662 pairs, 19,471 with thinking off). The languages disagree on 31.4% [30.7, 32.0] of pairs, and in 92% of these English is right and Hindi wrong. Mean RDS is 0.917 for disagreeing and 0.809 for agreeing pairs. Each 0.1 of RDS multiplies the odds of disagreement by 4.93 [4.58, 5.30], with predicted disagreement near zero below RDS 0.7 and climbing above 0.8 (Appendix Figure 6). Adding the gap in trace length raises the OR to 5.58 [5.18, 6.03], so length differences alone do not appear to explain the link. RDS and correctness come from the same generations, so this is only an association.

The answers inside the score. RDS includes the final-answer tokens, so two traces that reach different answers already differ. In pairs where both languages are right, replacing only Hindi's answer values accounts for 68% of the RDS difference between agreeing and disagreeing pairs. Removing answer tokens lowers the OR per standard deviation from 6.47 to 1.65, and comparing only step types (`ADD` against `SUB`, without operands) gives 1.14 (Appendix Figure 5). Out of fold, step types add 0.0007 AUC [−0.0004, 0.0018] to a model that knows task, difficulty and distractor. Comparing the two final answers alone gives AUC 0.951. RDS may therefore say little about reasoning routes beyond the answers. The step-type view is coarse, however, and within a question a slope survives on date recurrence (OR per 0.1, 1.42 [1.25, 1.61]) and day of week (1.26 [1.11, 1.43]).

5.2 Thinking-On Set

With hidden thinking on, the gap almost disappears, though thinking and token budget change together. On the same 450 questions, the CoT gap falls from 27.3 points to 1.0 (answered), or from 27.4 to 2.7 (strict).

Accuracy. Hindi CoT rises from 64.1% to 97.9% (strict 59.9% to 89.9%), and English CoT from 91.4% to 98.9%. Direct answers also rise: English from 64.2% to 97.9%, and Hindi from 59.2% to 94.9%.

Language. The hidden reasoning on Hindi CoT is 79% English words while the visible answer stays

in Hindi. This fits the en-CoT result without confirming it. Cut at 4,096 tokens, strict Hindi CoT would score 43.8%, below the 59.9% with thinking off, so the extra budget may matter as much as the language.

Format arguments. On Hindi direct prompts, a third of hidden steps (207,097 of 627,487) argue about the output format, against 0.2% of visible CoT steps. This may follow from the prompt, which asks for only the final answer, the whole response in Hindi, the word `Answer:` in English and Western digits, all at once. One trace spends 5,170 words on adding one day, and 361 of its 636 steps are `FORMAT`:

> "If I write: 2025-04-07. That's not in Hindi. … But if I have no text outside the last line, then there's no Hindi text."

The share is highest where the math is easiest: 49.8% of steps on date and time addition, against 15.3% on interval date. These arguments use up tokens but do not appear to make answers wrong: right traces have more `FORMAT` steps than wrong ones (157 against 120 on average).

Errors. Most errors in this set are unanswered responses: 287 of the 422 wrong ones have no answer line, mostly because they hit the 16,000-token limit. On matched questions with unanswered responses counted as wrong, disagreement falls from 31.7% to 10.9% (OR 0.23 [0.18, 0.30]).

5.3 en-CoT Set

Switching only the reasoning language to English accounts for most of the gap. On the same 2,250 questions, English CoT scores 88.9%, en-CoT 82.9% and Hindi CoT 59.4%. Of the 29.5-point gap, 23.5 points lie between en-CoT and Hindi CoT, where only the requested reasoning language changes. That is 79.7% of the gap, or 80.7% under strict scoring.

Accuracy. The remaining 6.0 points lie between the question languages, close to the 4.5-point gap under direct answering on these questions. The reasoning part is larger on every task and in every distractor condition (Figure 2).

Traces. English CoT and en-CoT traces, which share a reasoning language, are about as far apart (RDS 0.730) as two English traces of one question with and without a distractor (0.713). en-CoT and

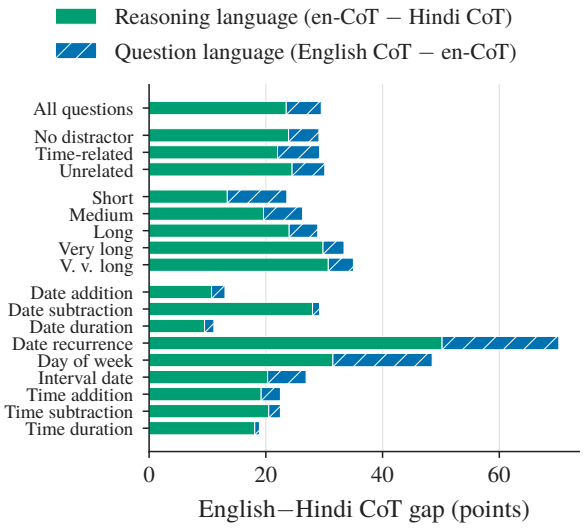


Figure 2: The English-Hindi CoT gap on 2,250 questions, split on one path through three conditions. No condition pairs English questions with Hindi reasoning, so the split is descriptive.

Hindi CoT traces on identical Hindi questions are about as far apart (0.837) as English and Hindi CoT (0.852), and traces of unrelated questions sit at 0.959 (Appendix Figure 7). A verification phrase appears in 86.1% of English CoT, 74.2% of en-CoT and 1.3% of Hindi CoT traces. An observational estimate attributes only 1.0% [−6.3, 9.0] of the gain to it.

Errors. The switch appears to remove errors tied to writing Hindi and to leave errors tied to reading it. en-CoT never walks forward and gives broken weeks about as often as English CoT (Table 6). It still counts backwards on recurrence in 28% of wrong answers, apparently reading आज को छोड़कर (*āj ko chorkar*, 'excluding today') as 'the occurrence before today'. These answers cluster at $k = 1$. About two-thirds of answers to Hindi $k = 1$ questions go backwards under en-CoT (67.1%), Hindi CoT (67.8%) and CLP (69.0%), against 3.5% for English CoT (Appendix Figure 4). The template's non-standard 1वीं (*1vīṃ*, '1st') is a candidate cause. We did not rerun a corrected template, and equally non-standard ordinals at $k = 2$ to 4 rarely do this. en-CoT is also wrong on 19% of "0 hours" questions, against 5% of others.

5.4 CLP Set

CLP helps, but less than one English turn. CLP scores 70.3%, 10.9 points above Hindi CoT and 12.6 below en-CoT. This reverses the order Qin et al. (2023) report for GPT-3.5 on MGSM, although

model, task and settings all differ.

Stage analysis. One candidate reason is that the understanding turn often solves the problem already and the solving turn copies it. On the three date-answer tasks, the second turn is right in 97.5% of responses when the first ends at the gold date, and in 20.1% when it does not. The first turn was not randomised, so this is not proof, and our temperature and language rule may also contribute.

5.5 Hinglish Set

Hinglish leaves Hindi questions unchanged on average and makes English ones worse. It costs English questions 18.0 points [16.8, 19.2] (strict 15.7), on all nine tasks and in each distractor condition (Appendix Figure 3). Hindi questions do not change on average (-0.0 [-1.3, 1.2]), so the gap narrows only because English falls.

Script. On Hindi questions, 68.5% of Hinglish traces are mostly Devanagari despite the Roman-script instruction, so this arm mostly tests code-mixed Devanagari. Traces that do use Roman script score no better.

Errors. The English loss seems to go beyond the cost of writing Hindi. English-question traces with no Hindi function words still lose 10.3 points [8.3, 12.3], and the share that verify falls from 86.5% to 18.6%. On date subtraction across a month boundary, English questions fall from 90.1% to 42.3%, near Hindi CoT's 36.2%, while same-month questions stay at 97.4%.

5.6 Trace Interventions

Removing the distractor helps a little, and we could not isolate the written mention. A mention may lead the model astray, or traces already going wrong may tend to talk about it.

Setup. We take wrong CoT responses to questions with a time-related distractor that mention it before answering: 526 Hindi and 37 English. Each experiment cuts the trace at or before the first mention, changes one thing, and lets the model continue (Appendix Table 7). The API always starts a new reply, so we served an FP8 build with vLLM. It scored 5.7 points above the API on 278 Hindi questions and 24.7 below on 22 English ones, so English results are descriptive only.

Deletion. On 222 Hindi traces, deleting the mention helps no more than deleting a neutral sentence (+0.0 [-6.8, 7.2]). With one continuation per version this test had about 24% power for a 5-point effect, so it does not show the effect is zero.

Resampling. Hand edits may understate effects (Macar et al., 2026), so, following Bogdan et al. (2025), we also regenerate seven continuations per trace from four starting points. In Hindi, continuing without the distractor in the question (Q) reaches 45.2% and with it (P) 40.0%, a difference of +5.3 points [3.2, 7.4] (Appendix Figure 8). This suggests the distractor was necessary for at least 8.8% of the errors made from that point (Tian and Pearl, 2000). Most continuations that go wrong with it also go wrong without it. The January sentence supplies 58% of the effect (without it, +2.8 [0.4, 5.0]), and the interval reaches zero once the 29 sentences are treated as a sample.

The mention. Keeping the sentence with the first mention lowers accuracy by 3.3 points [1.2, 5.4], but not detectably more than keeping the sentence before it ($p = 0.09$ on 248 traces). We cannot tell it apart from one more step along a failing trace. Free continuations also mention the distractor in 57.8% of rollouts, so Q against P changes the question and the writing at once.

6 Discussion

Across accuracy, trace distance, verification and error types, the requested language of reasoning seems to account for more of the gap than the language of the question, in line with prior work on English reasoning (Shi et al., 2023b; Yong et al., 2025; Qi et al., 2025). Method rankings did not transfer: one English turn beat CLP's two, and Hinglish lowered accuracy on English questions. Trace evidence also needs care, since an edit-distance score can pick up its signal from the answers.

7 Conclusion

On Sarvam-105B and TRD, CoT widens the English-Hindi gap from about 4 points to about 29. On the same Hindi questions, English reasoning closes most of it, CLP about a third, and Hinglish none. Hidden thinking nearly closes it at four times the token budget. The Hindi losses appear tied to methods seen mainly when writing Hindi. Trace divergence goes with disagreement, much of it through the answers, and we could not show that distractors in traces cause errors.

Limitations

Scope. Everything here concerns one model, greedy decoding except in the resampling arms, one prompt per condition, one dataset seed and two languages. Intervals cover question sampling only, so they say nothing about other wordings.

Causal claims. Language and prompt wording are confounded in every comparison across prompting conditions. No condition pairs English questions with Hindi reasoning, so our split is one path and depends on its order. Trace style, mentions and verification are chosen by the model or measured after it responds, so they support association only. Runs span about a month, the provider changed its API responses in that time, and condition and run date are partly confounded.

Data. The added conditions use the first 50 of 150 questions per cell, too few for distractor contrasts of a few points. Hindi distractors are not literal translations, and some Hindi question templates are awkward. Distractor sentences are a fixed pool assigned unevenly across tasks. TRD's time-task offsets in the code differ from the ranges its paper lists.

Measurement. The parser is regex-based and imperfect, RDS depends on it and on the edit-distance costs, and RQ5 drops pairs with unparseable answers. `FORMAT` cues are broad and can catch ordinary uses of words such as "English". Mention and verification rules are lexical. The error labels come from a script, leave about two-thirds of wrong standard responses unlabelled, and their percentages were not checked by hand. All hand-labelled samples were read by a single LLM annotator without human validation, including parser precision and our review of the Hindi templates.

Interventions. Both experiments cover only originally wrong traces and time-related distractors, on a quantised self-hosted model, and resampling uses a different temperature from the source traces. Unanswered Hindi rollouts rise from 5.7% to 10.0% with the distractor there, which the API data does not show, so part of the effect may belong to this setting. The resampling design was first committed after the run and was not publicly registered. We did not test whether mentions harm traces that would otherwise be correct.

Thinking mode. This set has 10 questions per cell and changes thinking and token budget together.

Future work. The next steps are: rerunning recurrence with corrected Hindi templates; adding English questions with Hindi reasoning; several paraphrases of each prompt, including a Hindi direct prompt with fewer format rules; CLP with solving forbidden in stage 1; thinking on and off at equal budgets; a question-by-mention factorial on correct and wrong traces; and human annotation of the labelled samples.

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Appendix A. Worked Examples

Hinglish, English question, wrong. Date subtraction, short. "Today is 2025-05-01, what was the date 3 days ago?" Gold 2025-04-28. Excerpt: *01 minus 3 karne par hum 31 April par aayenge* ('subtracting 3 from 01 brings us to 31 April'). Answer: 2025-04-30.

en-CoT, backward answer. Date recurrence, short, ID 5, on आज को छोड़कर, 1वीं आवृत्ति (*āj ko chorkar, 1vīṃ āvṛtti*, 'excluding today, the 1st occurrence'). Excerpt: *Given the phrasing "excluding today," this suggests we need to find what comes before.* Answer: 2025-04-06. Gold 2025-04-14.

CLP, stage 1 anchors stage 2. Date recurrence, long, ID 12. Stage 1 ends: *the date of the 7th occurrence is 2 August 2027.* Stage 2 ends: *Adding 78 days to 16 May 2027 also results in 2 August 2027.* Gold 2027-08-15.

Hindi CoT subtracts by walking forward. Date subtraction, very very long, ID 79. Today is 2027-05-09; what was the date 52 days ago? Gold 2027-03-18. Excerpt, translated: *From 9 May to 31 May is 22 days. … April has 30 days, so the date exactly 30 days before 30 April is 1 May.* Answer: 2027-05-01, after today.

Hindi CoT returns a "week" of two days. Interval date, long, ID 95. Today is 2030-05-15. Gold 2030-05-20 to 2030-05-26. Excerpt, translated: *15 May 2030 is a Saturday. … The next Monday will be 18 May 2030. … Sunday will be 19 May 2030 (6 days after Monday).* 15 May 2030 is a Wednesday.

English CoT is one day late. Day of week, very very long, ID 20, 2029-07-02. Gold Monday. Excerpt: *Total: 31 + 28 + 31 + 30 + 31 + 30 + 2 = 183 days.* Answer: Tuesday. 1 January to 2 July is 182 days.

English CoT skips 29 February. Date recurrence, medium, ID 60. Today is 2028-02-15, every 6 days, 4th occurrence. Gold 2028-03-10. Excerpt: *From 2028-02-27 to 2028-03-05: 28 - 27 = 1 day in February, plus 5 days in March = 6 days* ✓ Answer: 2028-03-11. That gap is 7 days, and the check repeats the mistake.

Appendix B. Further Statistics

Standard set. The Hindi direct-to-CoT penalty for time-related sentences has interval [0.6, 3.8], and removing any one sentence leaves it between 1.8 and 2.6. The Hindi-minus-English penalty is +2.6 [0.4, 4.7] answered but +1.4 [−0.7, 3.5] strict. Removing the January sentence takes the Hindi recurrence penalty from +4.5 to +0.3 [−4.6, 4.9]; English stays at +8.5. Of the 92 cut-off traces under that sentence, none is a verbatim loop. Under strict scoring the CoT gap grows from 23.0 to 31.1 with difficulty.

RDS. Of 21,600 loaded pairs, 938 (4.3%) are dropped because an answer could not be parsed, leaving 20,662 on 6,739 questions. With thinking off, 11,309 pairs are both correct, 5,947 English only, 505 Hindi only and 1,710 both wrong. Disagreement rises from 26.7% on short questions to 38.6% on very very long ones. A substitution between a partial and a full date costs 0.5 in RDS. These are in-sample associations.

en-CoT and CLP. The 29.5-point gap has interval [28.1, 30.9], the reasoning part 23.5 [22.1, 24.9], the question part 6.0 [5.0, 7.1], and the reasoning share 79.7% [76.4, 82.9]. The two parts are not distinguishable at the short level. CLP gains +9.0, +10.4 and +13.2 over Hindi CoT by distractor condition. en-CoT beats CLP by 12.6 [11.4, 13.9]; on MGSM Qin et al. report CLP 70.6 and En-CoT 57.8.

Hinglish. On Hindi questions, Hinglish lowers date duration by 5.5 points and raises date subtraction by 5.1, which cancel in the mean. Roman traces do no better than Devanagari ones (−2.8 [−5.2, −0.5]).

Interventions. The net 8.8% has interval [5.5, 12.1], and resampling the 29 sentences widens the Q−P interval to [−0.3, 9.4]. The mention test is (M−P)−(P−P1) = −4.6 [−9.6, 0.6]. Agreement with the API on 90 calibration questions is −2.2 points [−13.3, 7.8], which does not establish that the two are the same.

Appendix C. Secondary Figures and Tables

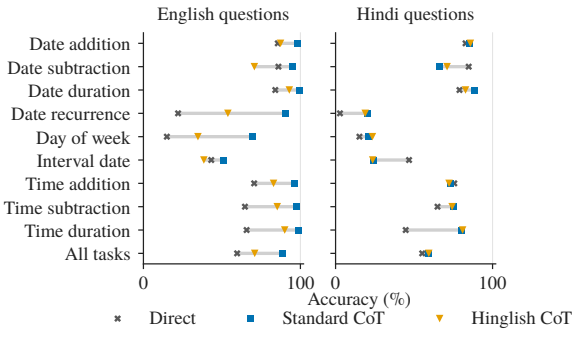


Figure 3: Accuracy by task under the Hinglish prompt, answered. Standard CoT reasons in the question's language.

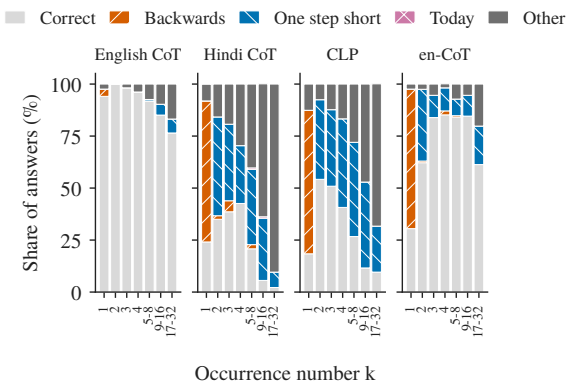


Figure 4: Date recurrence answers by occurrence number k, on 2,250 questions. Backward answers concentrate at $k = 1$ on Hindi questions; one-step-short answers run through every k.

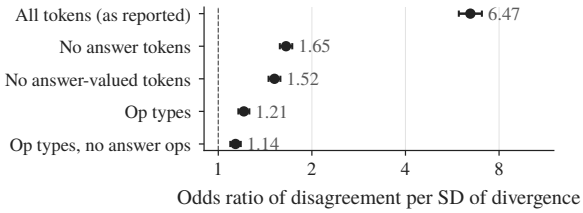


Figure 5: OR for disagreement per standard deviation of divergence, by what the score compares. Removing answer tokens takes most of the link away.

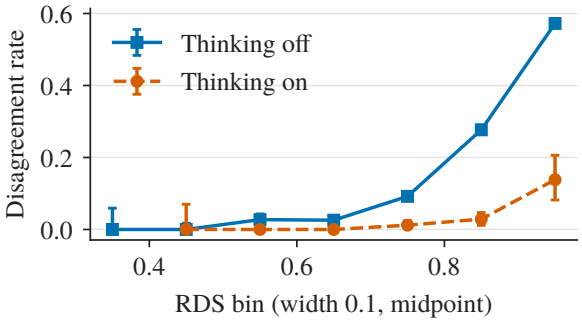


Figure 6: Observed disagreement rate by RDS bin of width 0.1, unadjusted, with 95% intervals; bins with fewer than 20 pairs are hidden.

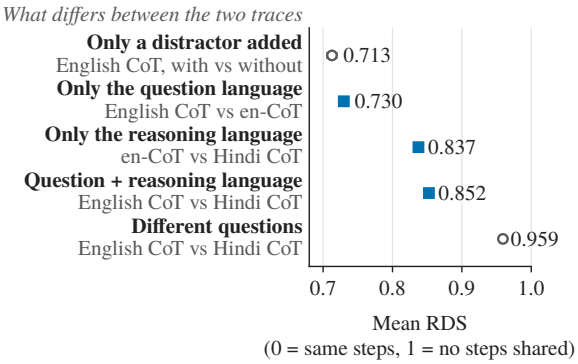


Figure 7: How different two traces are, as mean RDS with 95% intervals on 2,250 questions. Each row compares two traces that differ only in the factor in bold. Hollow grey rows are reference points: adding a distractor, and traces of unrelated questions.

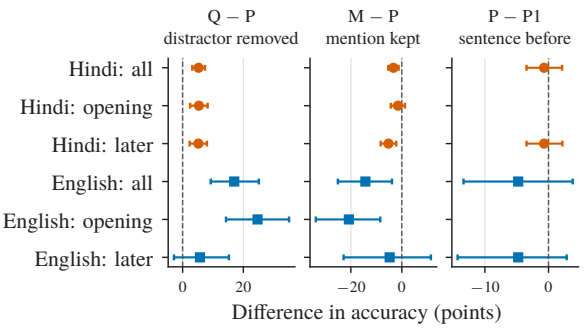


Figure 8: Resampling contrasts with 95% intervals. Q is the continuation without the distractor in the question, P with it, M after the mention, and P1 one sentence earlier. *Opening* traces mention the distractor in their first sentence and *later* traces after it. English is descriptive.

Variant	Prefix	Correct
Deletion: one greedy continuation		
A as written	s0 to s3	1 of 1
B mention deleted	s0 to s2	1 of 1
C neutral deleted	s1 to s3	1 of 1
Resampling: seven at temperature 0.8		
P before the mention	s0 to s2	6 of 7
Q P, distractor removed	s0 to s2	7 of 7
M after the mention	s0 to s3	7 of 7
P1 a sentence earlier	s0 to s1	6 of 7

Table 7: Every variant on one real Hindi trace (long date addition, ID 11), where s3 is the mention.